

Assignment 3

Group 15

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1.

- Fully developed use case description for desk booking.

Use case name:	Desk booking
Scenario:	Book a desk
Triggering event:	When an employee needs to reserve a desk for a specific time
Brief description:	The actor books a desk by navigating the office floor plans and choosing a desk. The system should create a record of the desk booking. The desk should be associated with that employee for that specific day and time slot. A confirmation email should also be sent to the employee.
Actors:	Employee
Related use cases:	Desk configuration management
Stakeholders:	Managers, IT department
Preconditions:	- To utilize the desk booking feature, users need to be allowed and verified. - Desks that can be reserved in the office area must be available. Users are unable to continue with the booking procedure if every desk is either reserved or not available. - Only during the designated reservation window—which may have restrictions like a maximum booking duration or an advance booking period—can users reserve workstations. - If possible, the system ought to take into consideration the preferences or requirements that users may have for the kind of desk they want to reserve, such as location, amenities, and seating configurations.
Postconditions:	- The user is given a specific desk for the desired date and time upon completing the booking process. - The user receives a confirmation email including information about their desk reservation, including the desk location, booking period, and any further instructions.

	- To guarantee that no other user may book the booked desk during the allocated time, the system modifies the availability state of the desk to match its reservation status. - To keep them updated on the progress of their reservation, the user is contacted in the event that there are any changes made to the desk booking (such as cancellations or modifications). - For auditing purposes, a record of the desk booking transaction is kept in the system's database, complete with information about the desk ID, booking duration, booking timestamp, and user ID.	
Flow of activities:	Actor	System
	1. User log into their account by using their username and password.	1.1 The system prompts for date, time, and desk location from the user.
	2. The user chooses the date, time, and desk location for booking.	2.1 According to the user choice, the system displays a list of available desks that meet the specified requirements.
	3. The user selects a desk from available options based on preferences and enters the duration	3.1 Asks for user confirmation.

	of the booking. 4. Confirm the booking.	4.1 The system sends a confirmation email with booking details.
Exception conditions:	1.1 The exception occurs when the user enters incorrect login details. The system verifies them against stored credentials; if no match is found, it prompts the user to input the correct credentials. 3.1 The exception arises when the user inputs an invalid booking duration, conflicting with system rules. The system verifies details and prompts the user to adjust the duration if necessary.	

- Fully developed use case description for desk configuration management

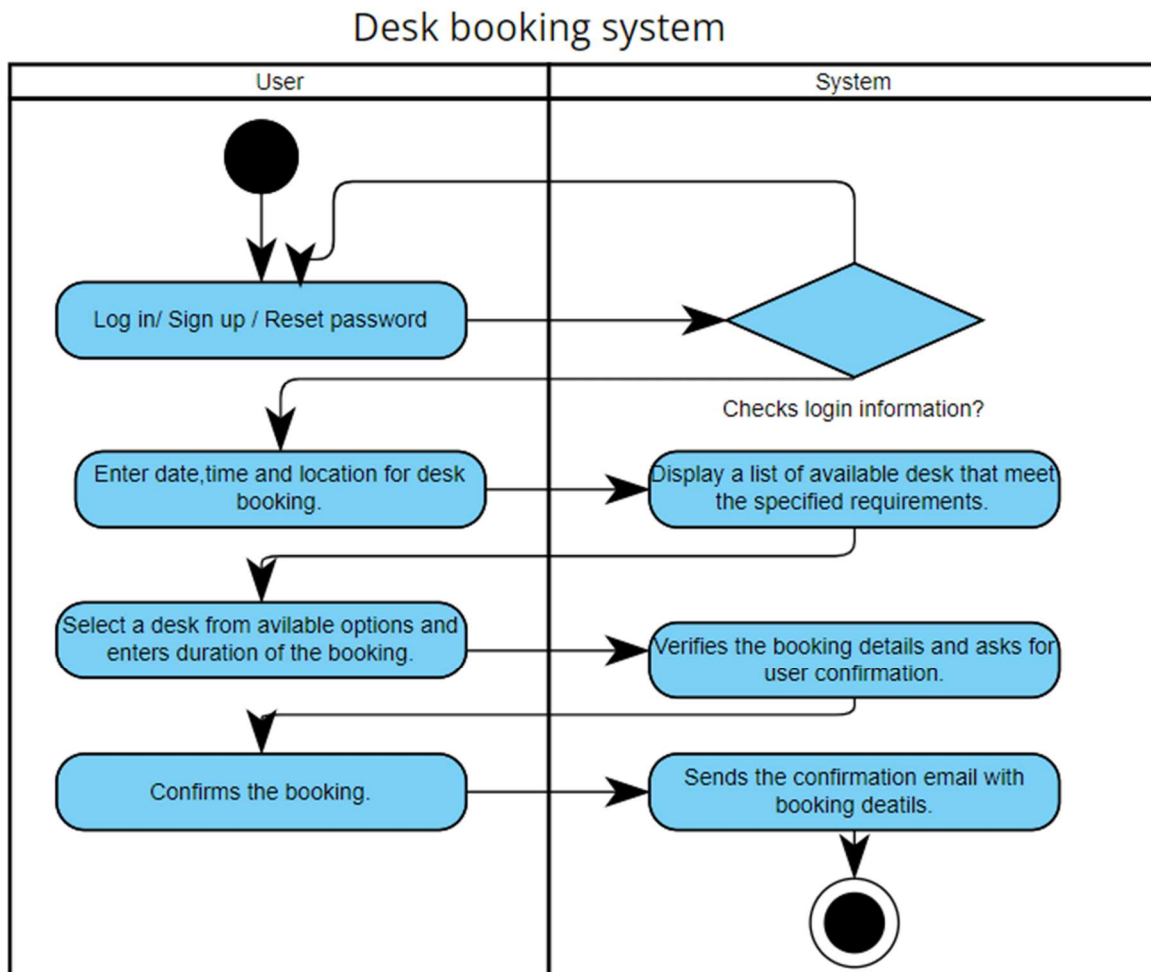
Use case name:	Desk configuration management
Scenario:	Desk configuration update
Triggering event:	When there is a need to set up, modify, or optimize the layout and arrangement of desks within the office space like office expansion, departmental changes, or introduction to new technologies, etc.
Brief description:	The actor books a desk by navigating the office floor plans and choosing a desk. The system should create a record of the desk booking. The desk should be associated with that employee for that specific day and time slot. A confirmation email should also be sent to the employee.
Actors:	Office Administrators
Related use cases:	None
Stakeholders:	Office Administrators, Facilities Manager
Preconditions:	- The required desk layouts must be able to be accommodated in the given office space. - Desks and related furniture should be included so that they can be configured and assigned.

	- To help with desk layout options, a thorough understanding of organizational needs, departmental hierarchies, and staff preferences is necessary. - Resources available for desk upkeep, repair, and reconfiguration as required.				
Postconditions:	- Effective desk configuration and allocation are made to satisfy organizational requirements, make the most use of available space, and take into account employee preferences. - Workspaces that are ergonomic and useful for employees' well-being and productivity are available to them. - To continuously increase workplace satisfaction, employee feedback on desk layouts is gathered and taken into consideration when making future configuration selections. - Desk upkeep and maintenance are checked to make sure they stay in good shape and to quickly resolve any problems.				
Flow of activities:	<table><tr><th>Actor</th><th>System</th></tr><tr><td> 1. Administrator login into desk configuration interface. 2. Administrators have the option to add, edit, or remove desks. </td><td> 2.1 After making any changes by the administrator system validate the changes. 2.2 The system checks if the changes are valid. 2.3 If the changes are valid then it applied to the database and reflected in the desk configuration. 2.4 If the changes are not valid then the administrator again adds, edits, or removes the desk. </td></tr></table>	Actor	System	1. Administrator login into desk configuration interface. 2. Administrators have the option to add, edit, or remove desks.	2.1 After making any changes by the administrator system validate the changes. 2.2 The system checks if the changes are valid. 2.3 If the changes are valid then it applied to the database and reflected in the desk configuration. 2.4 If the changes are not valid then the administrator again adds, edits, or removes the desk.
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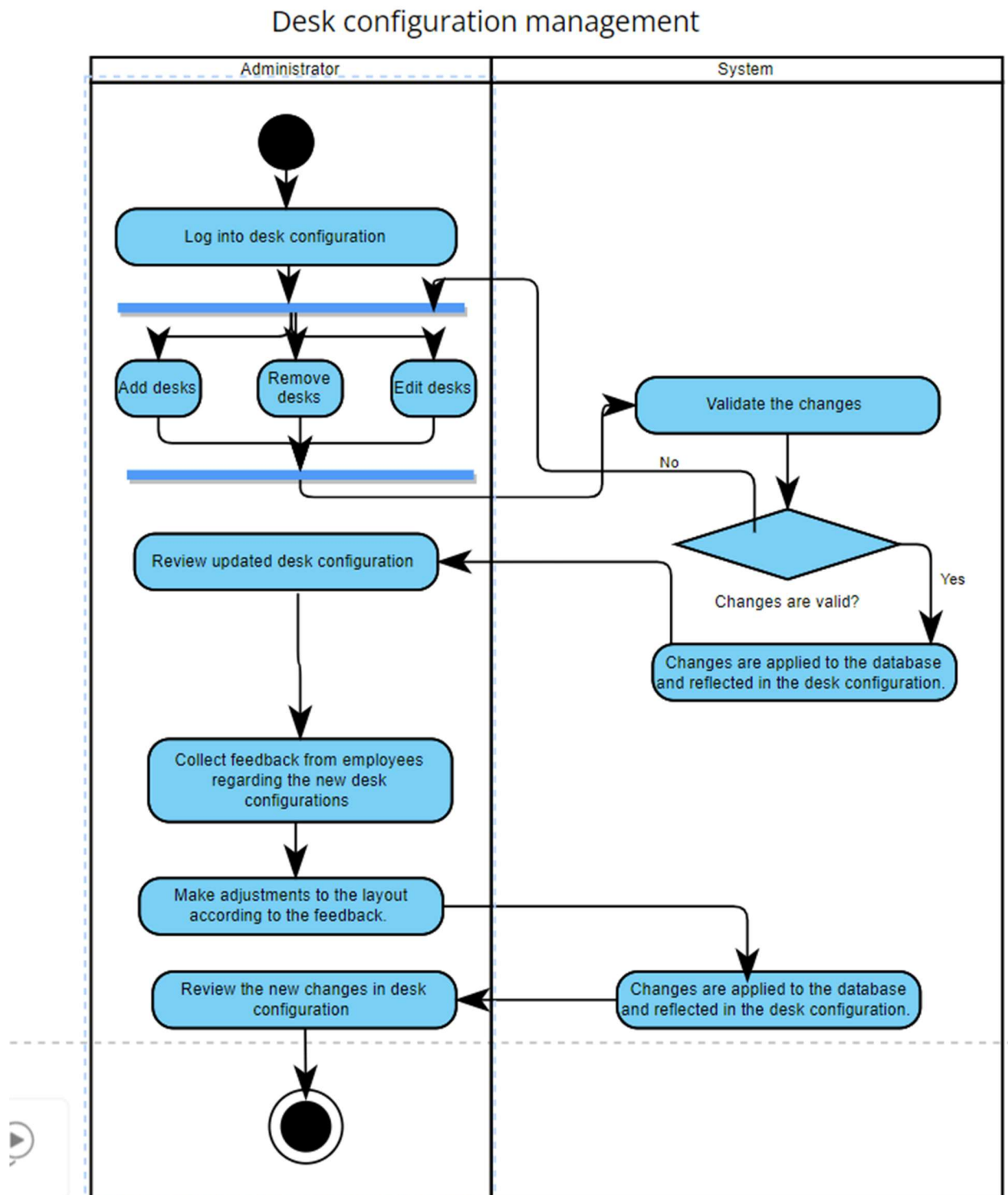
	3. The administrator reviews the updated desk configurations. 4. Administrators collect feedback from employees regarding the new desk configurations. 5. Make adjustments to the layout according to the feedback. 6. The administrator reviews the updated desk configurations. 5.1 The changes are applied by the system to the database and reflected in the desk configuration.
Exception conditions:	1.1 If admin credentials are incorrect, the system prompts re-entry or password reset. 3.1 If the database update fails, the system alerts the admin, offering troubleshooting guidance or support contact. 5.1 If layout adjustments violate policies (e.g., block emergency exits), the system flags it as invalid, prompting revision. 6.1 If the database update fails, the system alerts admin, offering troubleshooting guidance or support contact

Q2

- Activity diagram for desk booking

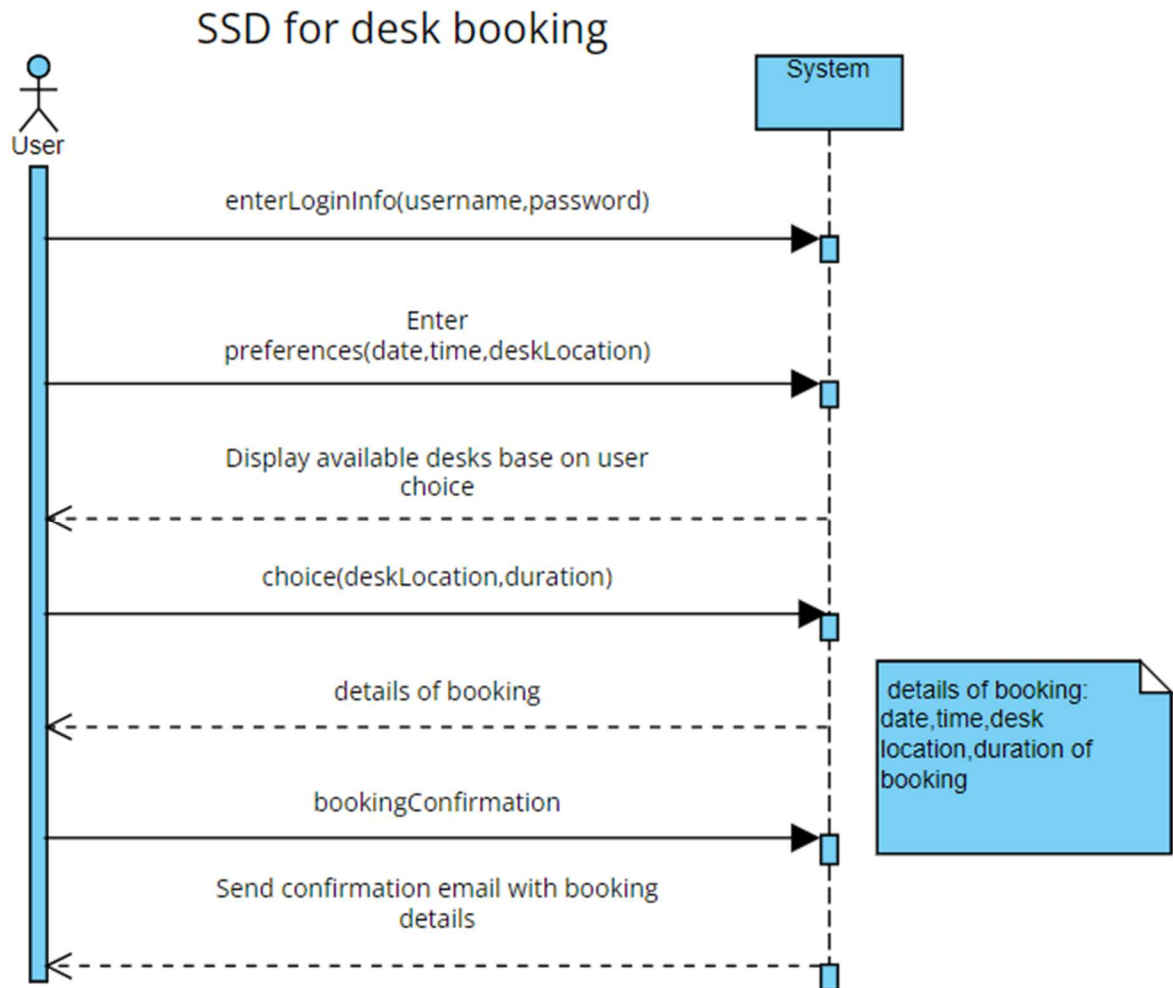


- Activity diagram for desk configuration management

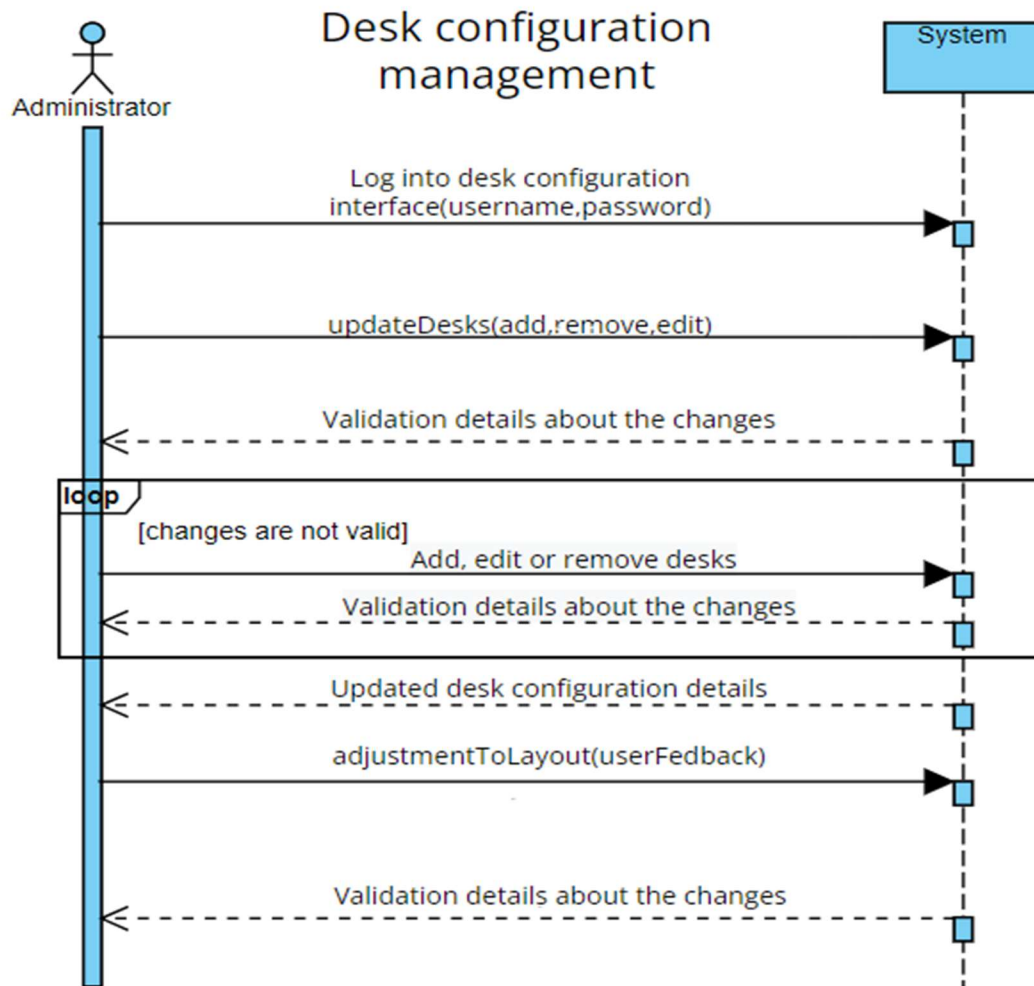


Q3.

- SSD for desk booking

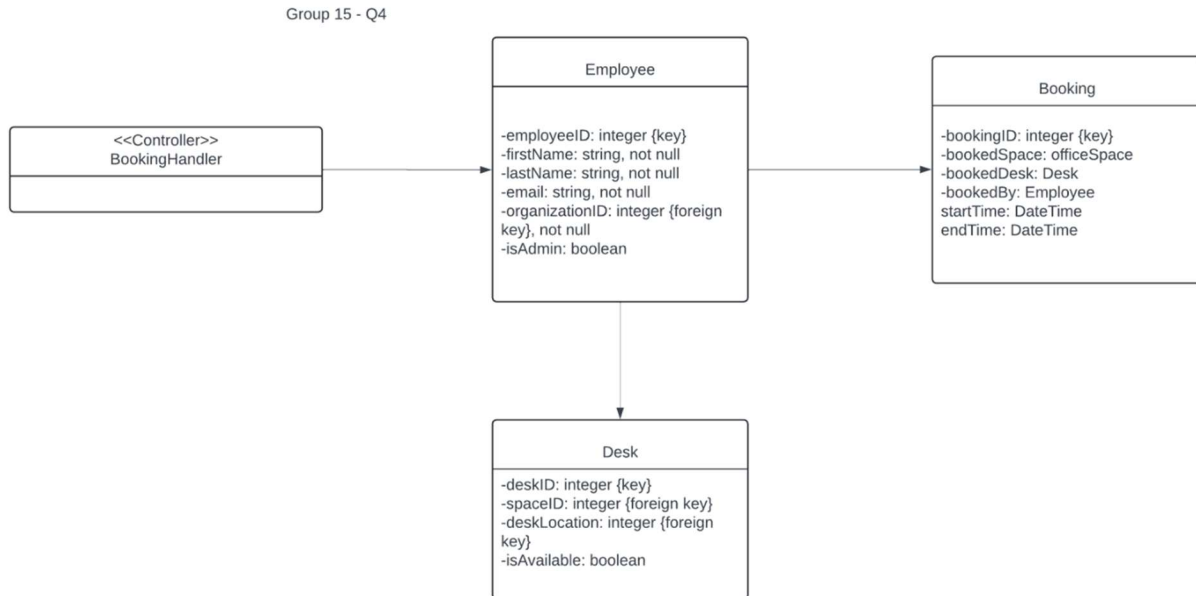


- SSD for desk configuration

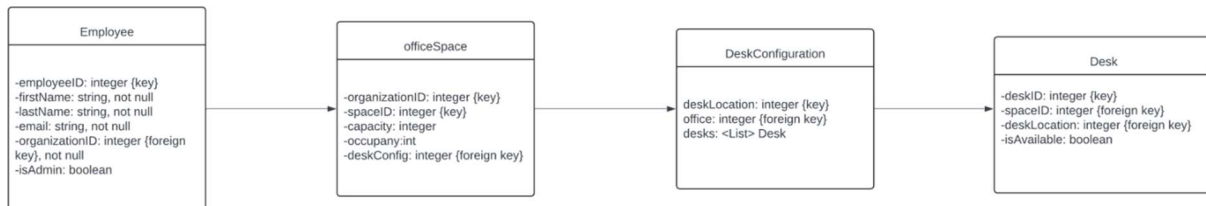


Q4. First Cut Design Class Diagram

a. Booking System



b. Desk Configuration



Q5. Agile CRC Cards and Sequence/Communication Diagrams

a. CRC Cards

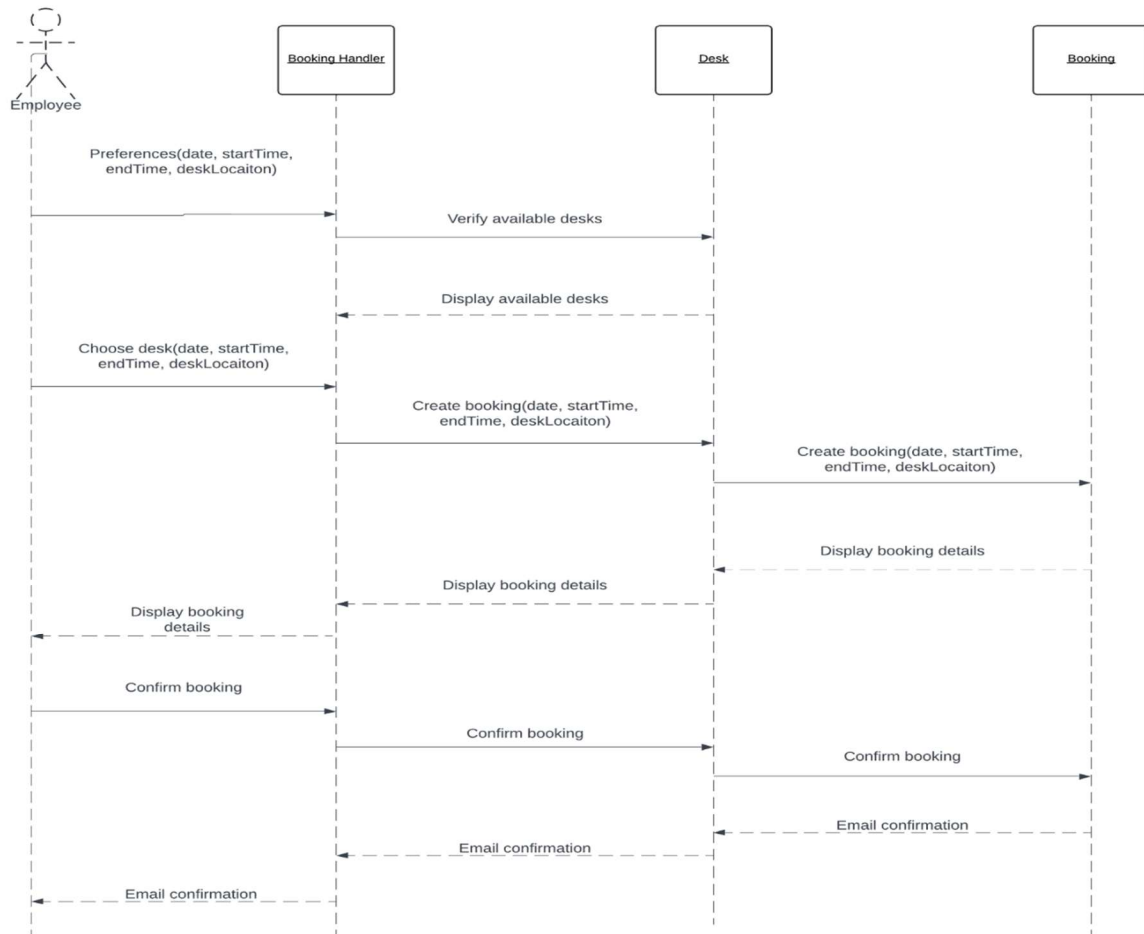
Employee (Front)	
Book space Cancel booking Send message Upload a file	Office Space Booking

Employee (Back)	
EmployeeID FirstName LastName Email OrganizationID IsAdmin	

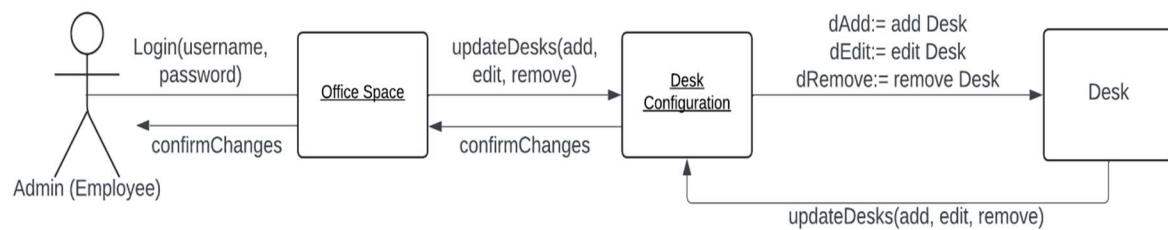
Booking (Front)	
Process booking Update booking Request booking history	Office Space Employee Desk

Booking (Back)	
BookingID Booked space Booked desk Booked by Start time End time	

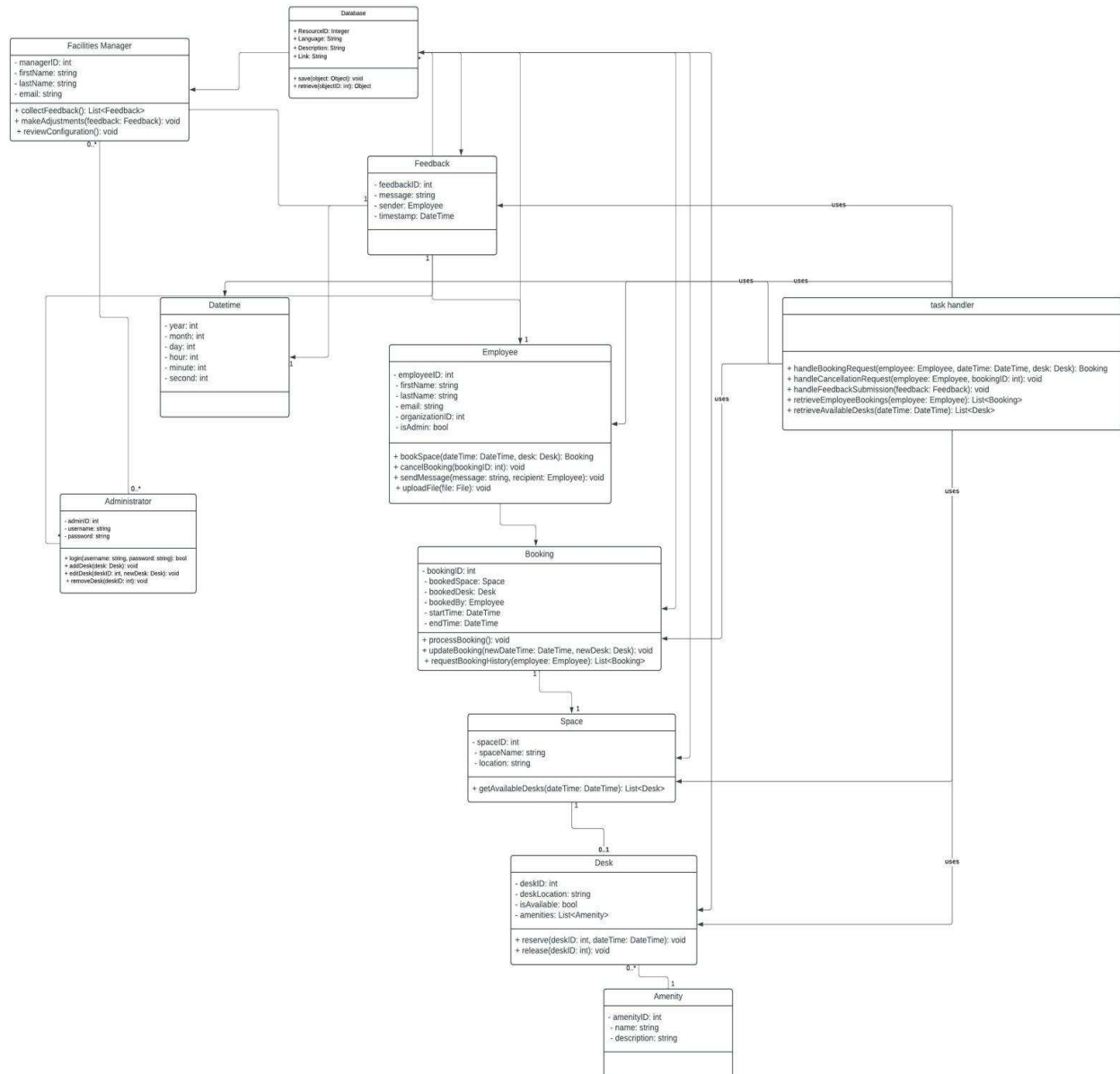
b. Sequence Diagram



c. Communication Diagram



Q6) Agile Fully Developed Design Class Diagram



Q7) Agile Class Package Diagram

